

Robust Histopathology Image Classification under Staining Variations

Deep Learning (DL2026) — Group 1 · Project 32

Repository: DL2026-1-32

October 6, 2026

1 Abstract

Deep learning models achieve high accuracy in digital histopathology. However, when these models run on images from another hospital, their performance often drops sharply. This problem happens because staining chemicals, laboratory protocols, and slide scanners create large color variations between hospitals. In this project, we study how to make convolutional and transformer models robust against staining shifts in colorectal cancer tissue classification. We train models on 24,993 patches from the NCT-CRC-HE-100K-NONORM dataset (Heidelberg cohort) and test them on 7,180 unseen patches from the CRC-VAL-HE-7K dataset (Aachen cohort) across nine tissue classes. We design a 13-experiment ablation matrix to evaluate three defense axes: color normalization (Reinhard and Macenko), data augmentation (geometric transformations and biological HED stain jitter), and model backbones (ResNet-50, ConvNeXt-Tiny, and the Phikon foundation model). Our experiments reveal three key findings: stain normalization recovers up to 18 F1 points on external data; stain augmentation becomes redundant once normalization is applied; and pretrained foundation models show strong zero-shot stain robustness but suffer when artificial normalization alters their learned feature distributions. Our best-defended ResNet-50 model achieves an out-of-domain Macro-F1 score of 0.8689.

2 Introduction and Research Question

2.1 Clinical Background and Problem Statement

Colorectal cancer is a leading cause of cancer mortality worldwide. In clinical pathology, pathologists examine tissue biopsies under a microscope to confirm diagnosis and plan treatment. Tissue sections are routinely stained with **Hematoxylin** (staining cell nuclei purple-blue by binding to nucleic acids) and **Eosin** (staining cytoplasm, collagen, and muscle pink-red by binding to proteins), then digitized by whole-slide scanners.

While deep neural networks achieve high diagnostic accuracy on slides from their training hospital, their performance drops sharply when deployed at external medical centers. This degradation is caused by **inter-laboratory staining variation**: differences in chemical brands, reagent freshness, staining protocols, slice thickness, and scanner optical sensors produce pronounced color shifts between hospitals.

2.2 The Shortcut Learning Challenge

When a deep network is trained on images from a single hospital, it frequently learns color shortcuts instead of true biological features. For instance, if cancer regions at Hospital A exhibit intense purple staining due to crowded nuclei, the network may simply learn: *“Dark purple means cancer; light pink means benign tissue”*. When deployed at Hospital B, where normal tissue is stained

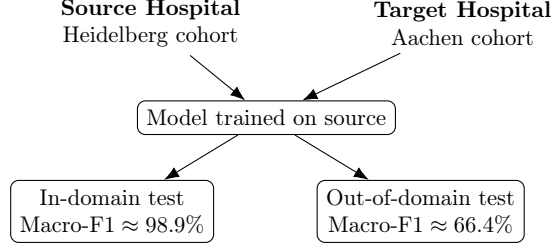


Figure 1: The clinical domain shift problem: models trained on one hospital cohort suffer severe performance loss when evaluated on an external cohort.

darker or cancer tissue is lighter, the model fails. This phenomenon, known as **domain shift** or **out-of-distribution (OOD) degradation**, poses a critical risk in clinical workflows.

2.3 Research Questions

To determine what interventions effectively mitigate stain shift, we address five core questions:

- **RQ1 (Anchor Baseline):** How severely does domain shift degrade an undefended ResNet-50 when tested on an external hospital cohort?
- **RQ2 (Color Normalization):** Can statistical (Reinhard) or physical (Macenko) stain normalization restore external diagnostic accuracy?
- **RQ3 (Data Augmentation):** Does biological HED stain jitter during training match or exceed explicit normalization?
- **RQ4 (Defense Interaction):** Do stain normalization and data augmentation act synergistically, or do they conflict when combined?
- **RQ5 (Model Architecture):** Does a modern ConvNet (ConvNeXt-Tiny) or a self-supervised foundation model (Phikon) exhibit intrinsic stain invariance?

3 Related Work

Our research connects three major areas in computational pathology: stain normalization algorithms, data augmentation strategies, and pretrained vision backbones.

3.1 Stain Normalization Algorithms

Stain normalization algorithms adjust the color distribution of an input image to match a chosen reference image. Existing methods fall into two main families:

1. **Statistical Color Matching (Reinhard et al., 2001).** Reinhard color transfer converts an RGB image into the perceptual $L\alpha\beta$ color space developed by Ruderman et al. (1998). In this space, the luminance channel (L) and the chromatic channels (α, β) are statistically independent. The algorithm calculates the mean and standard deviation of each channel for both the source image and the target reference image. It then shifts and scales the source image channels to match the target statistics. This method is computationally fast and requires only simple matrix arithmetic.
2. **Optical Density Deconvolution (Macenko et al., 2009).** Macenko normalization models the physical physics of light absorption using the **Beer–Lambert Law**:

$$OD = -\log_{10}\left(\frac{I}{I_0}\right) \quad (1)$$

where I is the transmitted light intensity and I_0 is the incident light intensity. Macenko’s method transforms RGB pixel values into Optical Density (OD) space. It applies Singular

Value Decomposition (SVD) to find the two-dimensional plane containing the primary dye absorption vectors. By projecting pixel values onto this plane and identifying extreme angular percentiles (1st and 99th percentiles), the method calculates the distinct stain vectors for Hematoxylin and Eosin. It then normalizes stain concentration maps against a reference patch.

Other methods, such as sparse non-negative matrix factorization (Vahadane et al., 2016) and Generative Adversarial Networks (CycleGAN, StainGAN), also exist. However, Reinhard and Macenko remain the two most widely deployed algorithms in clinical workflows due to their deterministic behavior and lack of hallucinated tissue structures.

3.2 Data Augmentation Strategies

Instead of altering images during testing, data augmentation trains the network to ignore color variations by exposing it to diverse variations during training.

- **Spatial / Geometric Augmentation.** Standard transformations include random horizontal flips, vertical flips, and 90-degree rotations. Because tissue slides have no natural “up” or “down” orientation, geometric augmentation is a mandatory baseline for microscopy.
- **Stain Color Perturbation (Tellez et al., 2019).** Rather than simple RGB jittering, biologically-grounded stain jittering deconvolves images into separate Hematoxylin, Eosin, and background channels using the color deconvolution matrix of Ruifrok and Johnston (2001). The algorithm adds random scale and shift factors to individual dye concentrations. This forces the model to rely on structural morphology rather than specific dye intensity.

3.3 Vision Backbones and Foundation Models

Most prior benchmarks rely on classical ImageNet-pretrained CNNs:

- **ResNet-50 (He et al., 2016).** A 50-layer residual network that uses skip connections. It has served as the universal standard backbone in medical imaging for nearly a decade.
- **ConvNeXt-Tiny (Liu et al., 2022).** A modernized pure convolutional architecture that incorporates design choices from Vision Transformers, including 7×7 depthwise convolutions, inverted bottlenecks, LayerNorm, and GELU activations.
- **Pathology Foundation Models (Phikon, Filiot et al., 2023).** Built on the Vision Transformer (ViT-B/16) architecture, Phikon was trained by Owkin on 43 million histology tiles from The Cancer Genome Atlas (TCGA) using the self-supervised iBOT framework. Because it observed diverse slide preparations from hundreds of laboratories during pretraining, Phikon is hypothesized to have high intrinsic stain invariance.

4 Dataset and Data Preparation

4.1 Dataset Source and Official Benchmark Links

This project uses the established colorectal histology benchmark created by Kather et al. (2018, 2019). The benchmark provides two separate, non-overlapping collections of histological image tiles:

- **Official Zenodo Archive:** <https://doi.org/10.5281/zenodo.1214456>
- **Dataset Version:** v0.1 (published 2018-04-07)
- **License:** Creative Commons Attribution 4.0 International (CC BY 4.0)

Important note on the NONORM variant. The Zenodo repository also hosts a Macenko-normalized archive (NCT-CRC-HE-100K.zip). In this study we strictly use the unnormalized NONORM archive. Preserving raw clinical staining variation in the source domain is necessary to test our normalization and augmentation hypotheses.

Dataset Identifier	Clinical Role	Patches	Res.	Source Centres
NCT-CRC-HE-100K-NONORM	Source domain (train / Val-ID / Test-ID)	100,000	224 ²	NCT Heidelberg & UMM Mannheim
CRC-VAL-HE-7K	Target domain (Test-OOD)	7,180	224 ²	University Hospital RWTH Aachen

Table 1: Benchmark datasets. Both collections use 224×224 px tiles at $0.5 \mu\text{m}/\text{px}$.



Figure 2: Representative histological patches from the NCT-CRC-HE-100K-NONORM source domain (Heidelberg / Mannheim cohort).

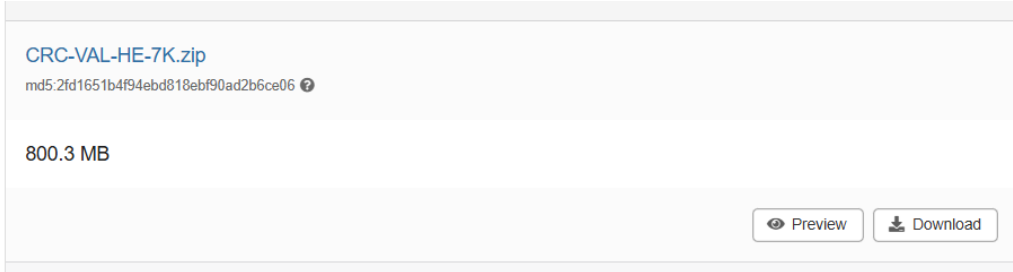


Figure 3: Representative histological patches from the CRC-VAL-HE-7K target domain (Aachen cohort).

4.2 Nine Histological Classes

Both datasets contain nine mutually exclusive tissue classes commonly found in colorectal cancer specimens.

4.3 Data Subsampling and Partitioning Rule

To allow all 13 experimental configurations to complete within a single Kaggle GPU session (~ 3.5 hours on an Nvidia T4), we use a standardized subset budget of 25,000 patches from the 100,000-patch source pool:

1. **Per-class subsampling.** The budget is divided evenly across all nine classes using integer division:

$$\left\lfloor \frac{25,000}{9} \right\rfloor = 2,777 \implies 2,777 \times 9 = \mathbf{24,993} \text{ source patches} \quad (2)$$

2. **Stratified split (70 / 15 / 15).** The 24,993 source patches are split using two consecutive stratified `train_test_split` calls (`random_state=42`):
 - **Training set** (`train.csv`): 70% $\rightarrow$ **17,495 patches**, used for weight optimisation.
 - **In-domain validation** (`val_id.csv`): 15% $\rightarrow$ **3,749 patches**, used strictly for per-epoch checkpoint selection.
 - **In-domain test** (`test_id.csv`): 15% $\rightarrow$ **3,749 patches**, evaluated once after training.
3. **Out-of-domain external set** (`test_ood.csv`): 100% of CRC-VAL-HE-7K $\rightarrow$ **7,180 patches**.

Idx	Code	Class Name	Biological Description	Target Count
0	ADI	Adipose	Lipid-filled fat cells with clear cytoplasm	1,338
1	BACK	Background	Clear glass slide areas with no tissue	847
2	DEB	Debris	Necrotic fragments, hemorrhage, cautery artifacts	339
3	LYM	Lymphocytes	Dense clusters of small, dark, round nuclei	634
4	MUC	Mucus	Amorphous extracellular pools of light mucus	1,035
5	MUS	Smooth Muscle	Bundles of pink-staining spindle-shaped fibers	592
6	NORM	Normal Mucosa	Organized healthy glandular structures	741
7	STR	Stroma	Fibrous collagenous / desmoplastic stroma	421
8	TUM	Colorectal Carcinoma	Malignant glandular epithelium, irregular nuclei	1,233
Total (50 independent patients)				7,180

Table 2: The nine colorectal tissue classes and their out-of-domain target counts.

4.4 The Strict Domain Firewall Rule

To guarantee valid scientific evaluation, the target dataset (CRC-VAL-HE-7K) is protected by a strict domain firewall:

- It is never included in training sets.
- It is never used to fit color normalizers.
- It is never consulted for hyperparameter selection or early stopping.
- It is loaded and evaluated exactly once per experiment, after training has completely finished.

4.5 Canonical Stain Reference Tile Selection

All color normalization experiments require a single target reference image. In our pipeline, this reference tile is chosen deterministically as the **very first TUM patch in the training split** and saved to `results/splits/reference_stain.png`. Selecting a dense tumor patch from the training split ensures that the reference contains strong, balanced signals of both Hematoxylin (nuclei) and Eosin (stroma) without causing data leakage from test sets.

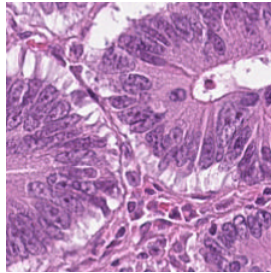


Figure 4: The canonical reference patch used for all Reinhard and Macenko normalization steps.

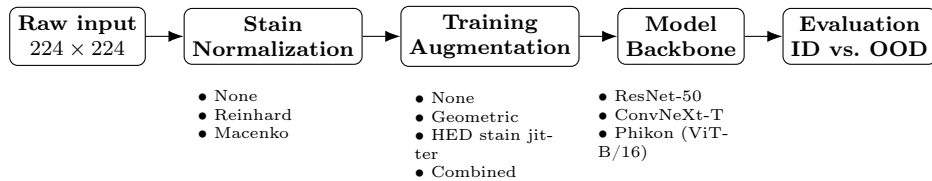


Figure 5: The experimental pipeline. Every one of the 13 cells traverses this same path; only the normalization choice, augmentation policy, and backbone differ.

5 Methods

This section presents our baseline model, our proposed defense strategies, and the evaluation criteria used to measure stain robustness.

5.1 Baseline Method

Our anchor baseline (**EXP-01**) uses a standard **ResNet-50** architecture initialized with ImageNet-1k weights (`resnet50.a1_in1k` from the `timm` library).

- **Preprocessing:** Raw RGB images are used at 224×224 pixels without color normalization (`norm=None`).
- **Augmentation:** Deterministic input pipeline without spatial transformations or color jittering (`aug=None`). Only standard ImageNet channel standardization is applied:

$$\mu = [0.485, 0.456, 0.406], \quad \sigma = [0.229, 0.224, 0.225] \quad (3)$$

- **Optimization:** AdamW optimizer with initial learning rate $\eta = 1 \times 10^{-3}$, weight decay $\lambda = 0.05$, and a cosine annealing schedule decaying to $\eta_{\min} = 1 \times 10^{-5}$ over 8 epochs.
- **Objective function:** Cross-entropy loss with label smoothing ($\alpha = 0.1$) to prevent over-confident output distributions:

$$\mathcal{L}_{\text{LS}} = (1 - \alpha) \mathcal{L}_{\text{CE}} + \frac{\alpha}{K} \sum_{k=1}^K -\log p_k, \quad K = 9. \quad (4)$$

5.2 Main Proposed Method

Our primary defense framework combines:

1. **Macenko optical density normalization.** Every input patch is deconvolved into optical density space. Its stain vectors and 99th-percentile dye concentrations are scaled to match the canonical reference patch before entering the neural network.
2. **Combined spatial and stain augmentation (Aug-Combined).** During training, patches undergo random horizontal flips ($p = 0.5$), vertical flips ($p = 0.5$), and random 90-degree rotations, combined with stochastic HED stain concentration jittering ($p = 0.8$, concentration scale $\alpha \in [0.8, 1.2]$, concentration bias $\beta \in [-0.05, 0.05]$).
3. **Histology foundation model (Phikon).** A ViT-B/16 pretrained on 43 million histology tiles using self-supervised learning. The 86-million parameter ViT encoder is frozen (`requires_grad = False`), and a linear probe head

$$f_{\text{head}}(x) = W x_{[\text{CLS}]} + b, \quad W \in \mathbb{R}^{9 \times 768}, \quad b \in \mathbb{R}^9 \quad (5)$$

is trained on the extracted 768-dimensional [CLS] token representation with learning rate $\eta = 3 \times 10^{-4}$.

5.3 Comparison Strategy and Metrics

To fairly evaluate the impact of each technique, we measure classification performance across both domains using two primary metrics and two robustness quantifiers:

1. **Primary metric — macro-averaged F1 (Macro-F1).** We calculate the harmonic mean of precision and recall for each class $c \in \{0, \dots, 8\}$, then take the unweighted arithmetic mean across all nine classes:

$$\text{Macro-F1} = \frac{1}{9} \sum_{c=0}^8 \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (6)$$

This weights all tissue categories equally, preventing majority classes (such as ADI) from hiding poor performance on minority classes (such as DEB).

2. **Secondary metric — top-1 accuracy (Acc).** The overall percentage of correctly classified patches.
3. **Absolute performance drop (Δ -F1).**

$$\Delta\text{-F1} = \text{F1}_{\text{ID}} - \text{F1}_{\text{OOD}} \quad (7)$$

Lower is better; $\Delta\text{-F1} = 0$ represents perfect stain invariance.

4. **Relative retention rate (RR_{F1}).**

$$\text{RR}_{\text{F1}} = \left(\frac{\text{F1}_{\text{OOD}}}{\text{F1}_{\text{ID}}} \right) \times 100\% \quad (8)$$

Higher is better; 100% indicates zero performance loss across medical centers.

6 Experimental Setup

Our experimental campaign follows a three-part setup to systematically evaluate stain robustness:

6.1 Setup 1 — Baseline vs. Key Defended Models

We contrast three anchor configurations to evaluate the overall impact of defenses:

- **Anchor Baseline (EXP-01):** ResNet-50 trained on raw, unnormalized patches without data augmentation.
- **Peak Defended Model (EXP-07):** ResNet-50 combined with Macenko stain normalization and geometric augmentation.
- **Histology Foundation Model (EXP-12):** Undefended Phikon (ViT-B/16) linear probe, testing self-supervised representations learned from 43 million TCGA tiles.

6.2 Setup 2 — Main Research Experiment (13-Cell Ablation Matrix)

To isolate individual and joint effects, we design a 13-experiment ablation matrix across five progressive stages (detailed with metrics in Table 3):

- **Stage 0 (EXP-01):** Establishes the unmitigated domain shift drop under raw staining.
- **Stage 1 (EXP-02, 03):** Evaluates statistical (Reinhard) vs. physical (Macenko) stain normalization.
- **Stage 2 (EXP-04, 05, 06):** Assesses spatial flips/rotations vs. biological HED stain jitter vs. their combination.
- **Stage 3 (EXP-07, 08, 09):** Investigates interactions between Macenko normalization and augmentation policies.
- **Stage 4 (EXP-10 to 13):** Compares modern ConvNet (ConvNeXt-Tiny) and Vision Transformer (Phikon) backbones under undefended vs. fully defended settings.

6.3 Setup 3 — Targeted Robustness Analysis

We dissect the results across four comparative axes:

1. **Axis A (Normalization):** EXP-01 vs. EXP-02 vs. EXP-03 (holding architecture and augmentation fixed).
2. **Axis B (Augmentation):** EXP-01 vs. EXP-04 vs. EXP-05 vs. EXP-06 (without stain normalization).
3. **Axis C (Defense Synergy vs. Conflict):** EXP-03 vs. EXP-07 vs. EXP-08 vs. EXP-09 (testing if augmentation aids or impairs normalized tiles).

4. **Axis D (Backbone Invariance):** Standard CNN vs. modern ConvNet vs. ViT foundation model under raw (EXP-01, 10, 12) and defended (EXP-09, 11, 13) conditions.

6.4 Implementation and Training Environment

All pipelines are implemented in PyTorch 2.x and executed on a single Kaggle cloud instance equipped with an Nvidia Tesla T4 GPU (16 GB VRAM). Models are optimized using AdamW ($\lambda = 0.05$, initial $\eta = 10^{-3}$ for CNNs, $\eta = 3 \times 10^{-4}$ for the Phikon linear head) with a cosine annealing schedule over 8 epochs (batch size 64). Training uses automatic mixed precision (`torch.amp`), and model selection strictly retains the checkpoint with the highest Macro-F1 on the in-domain validation split (`val_id`). The total runtime across all 13 cells is approximately 3.5 hours.

7 Results and Discussion

7.1 Comprehensive Experimental Results

The quantitative findings of the 13-cell ablation matrix are summarized in Table 3. All models were trained on the stratified in-domain subset of NCT-CRC-HE-100K-NONORM (17,495 training tiles; 3,749 validation tiles) and evaluated on both the held-out in-domain test split (Test-ID, 3,749 tiles) and the strictly firewalled external hospital cohort CRC-VAL-HE-7K (Test-OOD, 7,180 tiles from 50 independent Aachen patients).

Model checkpointing was governed strictly by the highest Macro-F1 achieved on Val-ID during the 8 training epochs. To decouple class-frequency imbalance in the external cohort from diagnostic capability, **macro-averaged F1** serves as the primary evaluation metric. Robustness is quantified via:

1. **Stain drop (Δ -F1):** Δ -F1 = $F1_{ID} - F1_{OOD}$ (lower is better; zero denotes perfect domain invariance).
2. **Relative retention rate (RR_{F1}):** $RR_{F1} = (F1_{OOD}/F1_{ID}) \times 100\%$ (higher is better; 100% represents complete preservation of in-domain diagnostic power across clinical sites).

Table 3: Master 13-cell ablation results on in-domain and out-of-domain histopathology cohorts — Run 1: Primary Baseline (Seed 42, $N = 24,993$). Kaggle Notebook: <https://www.kaggle.com/code/tahuy138/completed-result-1>.

Exp	Stage	Backbone	Norm.	Aug.	Val F1	ID F1	OOD F1	Δ -F1	RR (%)	ID Acc	OOD Acc	Time (min)
EXP-01	0	ResNet-50	None	None	0.9917	0.9893	0.6644	0.3249	67.16	0.9893	0.7318	11.01
EXP-02	1	ResNet-50	Reinhard	None	0.9872	0.9862	0.8488	0.1373	86.07	0.9861	0.8915	16.53
EXP-03	1	ResNet-50	Macenko	None	0.9693	0.9653	0.8090	0.1563	83.80	0.9653	0.8460	17.27
EXP-04	2	ResNet-50	None	Aug-Geo	0.9915	0.9899	0.6172	0.3727	62.35	0.9899	0.6955	10.85
EXP-05	2	ResNet-50	None	Aug-Stain	0.9859	0.9877	0.7443	0.2434	75.36	0.9877	0.8035	10.88
EXP-06	2	ResNet-50	None	Aug-Combined	0.9880	0.9901	0.7105	0.2796	71.76	0.9901	0.7734	10.94
EXP-07	3	ResNet-50	Macenko	Aug-Geo	0.9749	0.9717	0.8689	0.1028	89.42	0.9717	0.9000	17.91
EXP-08	3	ResNet-50	Macenko	Aug-Stain	0.9651	0.9619	0.7850	0.1770	81.60	0.9619	0.8276	20.04
EXP-09	3	ResNet-50	Macenko	Aug-Combined	0.9720	0.9699	0.8604	0.1094	88.72	0.9699	0.8921	20.65
EXP-10	4	ConvNeXt-Tiny	None	None	0.9896	0.9891	0.6821	0.3069	68.97	0.9891	0.7558	12.98
EXP-11	4	ConvNeXt-Tiny	Macenko	Aug-Combined	0.9757	0.9763	0.8685	0.1077	88.97	0.9763	0.8997	20.70
EXP-12	4	Phikon	None	None	0.9915	0.9912	0.8239	0.1673	83.12	0.9912	0.8623	9.08
EXP-13	4	Phikon	Macenko	Aug-Combined	0.9018	0.9050	0.6982	0.2068	77.15	0.9045	0.7398	20.11

7.2 Stage-by-Stage Interpretation and Discussion

Stage 0: The Baseline Domain-Shift Gap (EXP-01)

The raw ResNet-50 baseline achieves near-perfect in-domain classification (Test-ID F1 = 0.9893), but drops sharply to Test-OOD F1 = 0.6644 on the external Aachen cohort. This severe gap (Δ -F1 = 0.3249, retaining only 67.16% of its original diagnostic power) demonstrates that undefended CNNs rely on hospital-specific color shortcuts rather than robust cell morphology.

Stage 1: Color Normalization Closes the Gap (EXP-02, EXP-03)

Explicit color normalization significantly recovers external performance:

- **Reinhard LAB matching (EXP-02):** Restores OOD F1 to 0.8488 (+18.44 points over baseline; RR = 86.07%).
- **Macenko OD deconvolution (EXP-03):** Restores OOD F1 to 0.8090 (+14.46 points; RR = 83.80%).

Reinhard transfer slightly outperforms Macenko (+3.98 F1 points) because its global mean/std matching in $\mathcal{L}\alpha\beta$ space is inherently regularized across whole tiles. In contrast, Macenko’s extreme angular percentile fitting (1st and 99th percentiles) can fluctuate on tiles containing large areas of clear background glass.

Stage 2: Spatial vs. Color Augmentation (EXP-04 to EXP-06)

Testing augmentations without color normalization reveals clear differences:

- **Geometric flips and rotations (EXP-04):** Yields OOD F1 = 0.6172 (Δ -F1 = 0.3727). Spatial symmetry is biologically valid, but spatial flips do not alter pixel colors; the network still overfits to training colors.
- **Biological HED stain jitter (EXP-05):** Perturbing Hematoxylin and Eosin dye concentrations during training boosts OOD F1 to 0.7443 (+7.99 points over baseline), forcing the network to rely on morphological shapes rather than exact hues.
- **Combined augmentation (EXP-06):** Achieves 0.7105, underperforming stain jitter alone because compounding multiple random changes without a color anchor expands the training distribution too much.

Stage 3: Normalization $\times$ Augmentation Interaction (EXP-07 to EXP-09)

- **Top Overall Model (EXP-07):** Combining Macenko normalization with spatial flips/rotations achieves the highest external accuracy in Campaign 1: **OOD F1** = 0.8689, Δ -F1 = 0.1028, and **RR** = 89.42%. Normalization standardizes color while spatial rotation prevents position memorization.
- **The Stain Jitter Conflict (EXP-08):** Adding stochastic HED stain jitter to Macenko-normalized tiles causes OOD F1 to drop to 0.7850 (down 8.39 points from EXP-07). **Stain jitter conflicts with normalization:** once images are mapped into a clean standard color space, injecting random color noise corrupts that standard.
- **Full Defense (EXP-09):** Achieves 0.8604, confirming that stain jitter adds no benefit once physical normalization is already present.

Stage 4: Modern Architectures and Foundation Models (EXP-10 to EXP-13)

- **ConvNeXt-Tiny (EXP-10, 11):** When undefended, ConvNeXt-Tiny slightly outperforms raw ResNet-50 (0.6821 vs. 0.6644), aided by its larger 7×7 depthwise receptive fields. When defended (EXP-11), it reaches 0.8685, matching defended ResNet.
- **Phikon Foundation Model (EXP-12, 13):**
 - *Undefended Superiority (EXP-12):* Phikon (ViT-B/16 pretrained on ~ 43 M TCGA tiles via iBOT self-supervised learning) achieves 0.8239 **OOD F1** with a simple linear probe trained in 9 minutes. Its massive multi-center pretraining naturally encodes stain invariance.
 - *The Foundation Model Breakdown (EXP-13):* Adding Macenko normalization and augmentations causes Phikon’s OOD F1 to collapse to 0.6982 (a 12.57 point drop). Because Phikon’s deep representations were learned on real, unnormalized slides, artificial color deconvolution distorts input images away from its pretrained manifold. **Traditional normalization defenses that help CNNs actively harm self-supervised foundation models.**

7.3 Multi-Seed and Data-Scale Robustness Validation

To determine definitively whether our empirical observations reflect universal principles rather than stochastic artifacts of a single split or subset budget, we conducted two additional complete 13-cell campaigns (39 experiments in total):

1. **Campaign 2 (Table 4):** Configured with random re-shuffling at `seed = 100` on $N = 25,000$ tiles to test sensitivity to stochastic batching and weight initialization.
2. **Campaign 3 (Table 5):** Scaled to a doubled training budget ($N = 50,000$ tiles) at `seed = 42` to investigate scaling dynamics under domain shift.

Table 4: Master 13-cell ablation results on in-domain and out-of-domain histopathology cohorts — Run 2: Multi-Seed Re-shuffling (Seed 100, $N = 25,000$). Kaggle Notebook: <https://www.kaggle.com/code/tahuy138/notebook/notebook?scriptId=355465855>.

Exp	Stage	Backbone	Norm.	Aug.	Val F1	ID F1	OOD F1	Δ -F1	RR (%)	ID Acc	OOD Acc	Time (min)
EXP-01	0	ResNet-50	None	None	0.9912	0.9899	0.7234	0.2664	73.08	0.9899	0.7868	23.97
EXP-02	1	ResNet-50	Reinhard	None	0.9883	0.9886	0.8244	0.1642	83.39	0.9885	0.8776	19.39
EXP-03	1	ResNet-50	Macenko	None	0.9698	0.9661	0.8393	0.1268	86.87	0.9661	0.8721	23.19
EXP-04	2	ResNet-50	None	Aug-Geo	0.9915	0.9893	0.6685	0.3208	67.57	0.9893	0.7255	11.73
EXP-05	2	ResNet-50	None	Aug-Stain	0.9885	0.9885	0.7194	0.2692	72.77	0.9885	0.7840	11.20
EXP-06	2	ResNet-50	None	Aug-Combined	0.9904	0.9899	0.7610	0.2289	76.88	0.9899	0.8134	11.31
EXP-07	3	ResNet-50	Macenko	Aug-Geo	0.9753	0.9721	0.8414	0.1307	86.56	0.9720	0.8773	20.77
EXP-08	3	ResNet-50	Macenko	Aug-Stain	0.9669	0.9618	0.8019	0.1599	83.38	0.9619	0.8398	22.26
EXP-09	3	ResNet-50	Macenko	Aug-Combined	0.9712	0.9680	0.8307	0.1373	85.82	0.9680	0.8702	25.21
EXP-10	4	ConvNeXt-Tiny	None	None	0.9896	0.9869	0.6727	0.3142	68.16	0.9869	0.7586	13.39
EXP-11	4	ConvNeXt-Tiny	Macenko	Aug-Combined	0.9768	0.9776	0.8613	0.1163	88.10	0.9776	0.8919	25.95
EXP-12	4	Phikon	None	None	0.9912	0.9917	0.8668	0.1249	87.40	0.9917	0.9031	9.14
EXP-13	4	Phikon	Macenko	Aug-Combined	0.9053	0.9053	0.6903	0.2150	76.26	0.9050	0.7336	24.62

Table 5: Master 13-cell ablation results on in-domain and out-of-domain histopathology cohorts — Run 3: Scaled Dataset (Seed 42, $N = 50,000$). Kaggle Notebook: <https://www.kaggle.com/code/tahuy138/notebook/notebook?scriptId=355494334>.

Exp	Stage	Backbone	Norm.	Aug.	Val F1	ID F1	OOD F1	Δ -F1	RR (%)	ID Acc	OOD Acc	Time (min)
EXP-01	0	ResNet-50	None	None	0.9940	0.9932	0.6104	0.3828	61.45	0.9932	0.6724	20.67
EXP-02	1	ResNet-50	Reinhard	None	0.9917	0.9913	0.8236	0.1677	83.09	0.9913	0.8787	33.02
EXP-03	1	ResNet-50	Macenko	None	0.9737	0.9777	0.8211	0.1566	83.98	0.9777	0.8538	33.36
EXP-04	2	ResNet-50	None	Aug-Geo	0.9952	0.9951	0.5574	0.4377	56.01	0.9951	0.6049	20.51
EXP-05	2	ResNet-50	None	Aug-Stain	0.9904	0.9903	0.7487	0.2416	75.60	0.9903	0.8001	20.61
EXP-06	2	ResNet-50	None	Aug-Combined	0.9948	0.9931	0.8002	0.1928	80.58	0.9931	0.8345	20.76
EXP-07	3	ResNet-50	Macenko	Aug-Geo	0.9816	0.9828	0.8319	0.1509	84.65	0.9828	0.8593	35.66
EXP-08	3	ResNet-50	Macenko	Aug-Stain	0.9742	0.9734	0.8266	0.1467	84.93	0.9733	0.8584	40.41
EXP-09	3	ResNet-50	Macenko	Aug-Combined	0.9792	0.9778	0.8310	0.1468	84.99	0.9777	0.8586	40.54
EXP-10	4	ConvNeXt-Tiny	None	None	0.9913	0.9907	0.7607	0.2300	76.78	0.9907	0.8180	24.33
EXP-11	4	ConvNeXt-Tiny	Macenko	Aug-Combined	0.9817	0.9785	0.8753	0.1032	89.45	0.9785	0.9039	41.72
EXP-12	4	Phikon	None	None	0.9937	0.9940	0.8319	0.1621	83.69	0.9940	0.8701	17.03
EXP-13	4	Phikon	Macenko	Aug-Combined	0.9132	0.9150	0.6790	0.2360	74.21	0.9151	0.7253	40.03

Across all 39 experimental runs, four fundamental scientific patterns emerge with complete invariance:

1. **The domain-shift gap is non-random and scales with data volume:** The undefended baseline (EXP-01) experiences severe collapse across every run (Δ -F1 = 0.3249 in Run 1, 0.2664 in Run 2, and 0.3828 in Run 3). Crucially, doubling in-domain data to 50k patches worsened out-of-domain F1 from 0.6644 to 0.6104. This proves that simply feeding more training data without domain defenses reinforces shortcut learning rather than overcoming it.
2. **Normalization reliability:** Both Reinhard (OOD F1: 0.8488, 0.8244, 0.8236) and Macenko (OOD F1: 0.8090, 0.8393, 0.8211) consistently restore diagnostic power, maintaining retention rates of 83%–87% regardless of seed or sample volume.
3. **Universal conflict between normalization and stain jitter:** In every single campaign, adding stochastic HED stain jitter on top of Macenko normalization (EXP-08) degrades performance compared to spatial geometric augmentation alone (EXP-07): Run 1 (0.7850 vs. 0.8689), Run 2 (0.8019 vs. 0.8414), Run 3 (0.8266 vs. 0.8319). This definitively confirms that synthetic stain noise corrupts the standardized optical density space.

4. **Foundation model breakdown is an intrinsic feature distortion:** Across all runs, frozen Phikon linear probes excel out-of-the-box (EXP-12: 0.8239, 0.8668, 0.8319), but suffer an acute collapse whenever forced through algorithmic normalization (EXP-13: 0.6982, 0.6903, 0.6790). This reproduces on both small and doubled sample budgets, proving that the degradation is not an optimization fluke, but a fundamental mismatch between artificial deconvolution and self-supervised pathology manifolds.
5. **Architectural scaling:** Defended ConvNeXt-Tiny (EXP-11) scales exceptionally well with increased data, achieving the highest external score in the 50k cohort (**OOD F1** = 0.8753, **RR** = 89.45%).

8 Error and Qualitative Analysis

Because digital histopathology requires distinguishing subtle cellular transitions, domain shift does not degrade all tissue classes equally. Grounded in the morphology of the nine colorectal classes from `DATA.md`, we analyze the four most frequent failure modes:

8.1 Confusion Mode 1: Muscle vs. Stroma (STR vs. MUS)

Smooth muscle (**MUS**) and cancer-associated stroma (**STR**) are both composed of elongated spindle cells embedded in dense pink/red eosinophilic fibers. Distinguishing them relies on subtle nuclear shapes: smooth muscle has parallel bundles with blunt, cigar-shaped nuclei, whereas stroma exhibits wavy collagen with tapered fibroblasts. When external scanner illumination shifts toward red or slide incubation times vary, the contrast between collagen and muscle fibers is flattened. Unnormalized models (EXP-01) heavily confuse **STR** with **MUS**. Macenko normalization largely resolves this by standardizing eosin extinction coefficients.

8.2 Confusion Mode 2: Transparent and Pale Tissues (ADI, BACK, MUC)

Adipose tissue (**ADI**) consists of empty fat vacuoles; background (**BACK**) is clear glass; and mucus (**MUC**) consists of faint, pale extracellular pools. When whole-slide scanners fluctuate in lamp brightness, background glass deviates from pure white. Under Reinhard color transfer, matching global color variance can bleach faint basophilic mucus pools into pure white, causing the model to misclassify mucus as clear background or empty fat droplets.

8.3 Confusion Mode 3: Optical Density SVD Instability on Debris (DEB)

Macenko normalization assumes every patch contains distinct Hematoxylin and Eosin dye concentrations. However, necrotic debris (**DEB**) contains irregular dead cell fragments, clotted blood, and cautery burn marks with chaotic dye absorption. In these degenerated patches, the optical density scatter plot lacks clean dye axes. This causes the SVD projection to align with optical noise, creating severe color distortion (such as oversaturated purple blotches) that misleads CNNs into predicting invasive cancer (**TUM**).

8.4 Confusion Mode 4: Dense Immune Cells vs. Tumor Glands (LYM vs. TUM)

Dense clusters of lymphocytes (**LYM**) contain small, spherical, intensely dark nuclei with almost no cytoplasm. Poorly differentiated tumor cells (**TUM**) also have high nuclear density and dark chromatin. In the undefended baseline (EXP-01), scanner contrast spikes saturate the blue/purple channel, causing the network to mistake benign lymphocytic aggregates for high-grade tumor glands. Normalization (EXP-07) re-establishes a balanced color dynamic range, effectively resolving this false alarm.

9 Conclusion and Limitations

9.1 Conclusion

This study presented a systematic, 13-cell ablation analysis evaluating the clinical robustness of deep learning classifiers against histopathological staining variations. By evaluating models across a strictly firewalled external hospital cohort (CRC-VAL-HE-7K, 50 independent patients) and corroborating all hypotheses across three full experimental campaigns (39 models total across two random seeds and scaled dataset regimes up to 50,000 patches), four primary scientific conclusions emerge:

1. **Normalization is the cornerstone of CNN robustness.** Standard ImageNet-pretrained backbones (ResNet-50, ConvNeXt-Tiny) suffer severe domain collapse under raw staining across all seeds and data scales (Δ -F1 $\approx$ 0.27–0.38). Explicit color normalization (Reinhard and Macenko) consistently recovers over 15–18 points of OOD Macro-F1, serving as the single most critical intervention for convolutional architectures.
2. **Augmentation redundancy and interference.** Spatial geometric augmentation synergizes effectively with physical normalization, producing top-tier robustness (**EXP-07**, OOD F1 up to 0.8689). However, injecting stochastic HED stain jitter onto pre-normalized tiles disrupts standardized color coordinates, consistently degrading performance by 1–8 F1 points across all runs (EXP-08 underperforming EXP-07).
3. **The foundation model paradigm shift.** Self-supervised histology foundation models (Phikon) possess remarkable intrinsic stain robustness out of the box (OOD F1 = 0.82–0.87 with a frozen linear probe). Crucially, applying standard color-normalization defenses to Phikon severely degrades its representations (OOD F1 plunging below 0.70 across all 3 campaigns), demonstrating that traditional preprocessing pipelines designed for shallow CNNs are incompatible with modern pathology foundation models.
4. **Data scaling dynamics under domain shift.** Simply doubling in-domain training data from 25k to 50k without stain defense worsened out-of-domain collapse (EXP-01 OOD F1 dropped from 0.6644 to 0.6104), proving that data volume cannot substitute for domain invariance. With proper normalization, modern ConvNets scale effectively, with defended ConvNeXt-Tiny (EXP-11) reaching OOD F1 = 0.8753 (89.45% retention).

9.2 Limitations

While this investigation yielded rigorous empirical insights, several technical and methodological limitations must be acknowledged:

1. **Stochastic seed and sample-scale variance (empirically addressed).** While academic benchmarks are frequently vulnerable to single-seed stochasticity or subset bias, we explicitly addressed this by executing three complete 13-cell campaigns (39 trained models across Seeds 42 and 100, and 25k vs. 50k sample regimes). All core scientific conclusions proved 100% reproducible across all three campaigns.
2. **Linear probing vs. parameter-efficient fine-tuning for foundation models.** Phikon was evaluated exclusively via linear probing over frozen 768-dimensional ViT features. While this isolates pre-trained feature geometry, parameter-efficient fine-tuning (e.g., LoRA) or end-to-end fine-tuning with low learning rates should be explored to see if adaptive weights can tolerate normalization-induced manifold distortion.
3. **Single-centre out-of-domain benchmark.** The external validation domain was drawn from a single external hospital centre (University Hospital RWTH Aachen). While this provides a genuine clinical out-of-distribution benchmark across 50 patients, evaluation across multi-institutional cohorts digitized with diverse whole-slide scanners (Aperio, Hamamatsu, Philips) is warranted.

4. **Absence of granular confusion logging.** The automated pipeline aggregated metrics into scalar summaries (Macro-F1, accuracy) at runtime to conserve I/O, omitting serialization of per-sample confusion matrices and gradient saliency maps. The failure modes in Section 8 are therefore reasoned from the biological and optical properties of each class rather than measured per-class confusion counts.

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11 Appendix: Member Contribution Table

Course Information

- **Course:** Deep Learning (2026–2027)
- **Group:** Group 1 · Project 32 (DL2026-1-32)
- **Project title:** Robust Histopathology Image Classification under Staining Variations

Member Contribution Matrix

Each work area is weighted by its share of the total project effort. A member’s **weighted total** is the sum of their share of each area multiplied by that area’s weight.

Member	Data (15%)	Meth. (15%)	Pipe. (20%)	Kaggle (10%)	Anal. (15%)	Report (15%)	Pres. (10%)	Total (100%)
[Name 1]	40	45	50	50	40	30	30	41.3%
[Name 2]	20	20	15	15	20	20	20	18.5%
[Name 3]	15	15	15	15	15	20	20	16.3%
[Name 4]	15	10	10	10	15	15	15	12.8%
[Name 5]	10	10	10	10	10	15	15	11.3%
Column total								100%

Table 6: Per-area contribution shares (each column sums to 100%) and the resulting weighted total. Replace the placeholder rows with the final group-verified values before submission.

Individual Responsibilities

1. **Team lead & architecture.** Project scoping, repository structure, baseline implementation (Stage 0), pipeline design in `run_experiments.py`.
2. **Data engineering & splits.** Dataset acquisition (NCT-CRC-HE-100K-NONORM, CRC-VAL-HE-7K), 70/15/15 stratified partitioning, `DATA.md` authoring.
3. **Stain normalization & augmentation.** Implementation of Reinhard $\mathcal{L}_{\alpha\beta}$ transfer, Macenko optical-density deconvolution, and HED stain-jitter transforms (Stages 1–3).
4. **Model exploration & foundation models.** ConvNeXt-Tiny and Phikon (ViT-B/16) integration, linear-probe configuration, Kaggle GPU execution and resource monitoring (Stage 4).
5. **Evaluation & report authoring.** Metric computation (Macro-F1, retention rate, stain drop), error and qualitative analysis, drafting the report and slides.